Happiness Toolkit in a paediatric outpatient setting for children with social, emotional, behavioural and/or mental health difficulties: a prospective cohort study

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Resumo (Abstract)

Abstract Objective To evaluate the feasibility, acceptability and clinical outcomes of the Happiness Toolkit (HTK), a structured, low-intensity, family-delivered intervention for children aged 0–16 years screening positive for social, emotional, behavioural and/or mental health difficulties (SEMHD) in a paediatric outpatient setting. Design Single-site prospective cohort study; validated age-specific baseline screening, parent-reported outcomes at 6 weeks and 6 months. Setting General paediatric outpatient department, secondary hospital, Ireland. Participants Consecutive paediatric outpatient attendees screening positive for SEMHD on the Ages and Stages Questionnaires: Social-Emotional, Second Edition (under 6 years) or Screen for Child Anxiety Related Emotional Disorders questionnaire (6–16 years) were enrolled for intervention. Interventions The HTK comprises six evidence-based components practised for 5 minutes morning and evening, introduced during routine outpatient consultation. Main outcome measures Primary: consent rate, positive screen rate and 6-week response rate (feasibility); parental willingness to recommend (acceptability). Secondary: parent-reported functional changes at 6 weeks and 6 months, component uptake and adverse events. Results 240 of 277 families approached across 16 outpatient clinics consented (86.6%); 94 children screened positive for SEMHD (39.2%) and received the HTK. At 6-week follow-up (n=46; 48.9% response rate), 95.2% of parents reported a positive or very positive experience, 73.9% noted improved mood in their child, 43.5% reduced screen time and 37.0% improved sleep patterns. 97.8% would recommend, with no adverse events. At 6-month follow-up (n=13; 13.8% response rate), 61.5% reported sustained child mood improvement and 92.3% would still recommend. Conclusions This study demonstrates strong feasibility and high parental acceptability for the HTK when embedded in routine paediatric outpatient care. Parent-reported improvements at 6 weeks and no adverse events at both time points support progression to a controlled evaluation. In Ireland, 4462 children await Child and Adolescent Mental Health Services and over 28 000 await primary care psychology; the HTK represents a promising, pragmatic early and bridging support for emerging SEMHD.

WHAT THIS STUDY ADDS

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

Introduction

One in seven (14.3%) of 10–19-year-olds globally experience a mental health condition, with one-third of lifetime mental health conditions emerging before age 14 and half before age 18. 1 In Ireland, the burden is striking. A post-COVID population survey found that over 25% of Irish adolescents scored in the very high risk range on the Strengths and Difficulties Questionnaire, indicative of high risk for a diagnosable mental disorder. 2 Face-to-face structured diagnostic

interview of Irish 11–13-year-olds found that 27.4% met current criteria for a Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition (DSM-IV) Axis I mental disorder, one of the highest rates reported in any European paediatric population. 3 4 Severe anxiety in Irish children aged 12–19 doubled from 11% to 22% between 2012 and 2019. 5 This paper uses the term social, emotional, behavioural and/or mental health difficulties (SEMHD) for the range of non-neurodevelopmental psychosocial difficulties in childhood, consistent with Irish and UK health policy. Child and Adolescent Mental Health Services (CAMHS) is the publicly funded specialist child psychiatric service in Ireland and the UK.

Ireland faces particular challenges, where almost one in three young people have experienced elevated levels of depression and anxiety. 5 As of December 2025, 4462 children were on the CAMHS waiting list and over 28 000 awaiting primary care psychology, with delays of up to 10 years for primary care psychology in some areas. 6 7 Prolonged waits are associated with symptom escalation and crisis presentations. Clinical judgement alone has low sensitivity for identifying behavioural and emotional problems in paediatric primary care (14% to 54%), whereas validated screening tools recommended for routine use achieve sensitivities of 70% or higher. 8

Paediatric outpatient clinics are a critical but underutilised point of early intervention. Despite American Academy of Pediatrics guidance recommending periodic validated psychosocial screening, 8 Ireland has no national programme in paediatric care. Family-led, low-intensity interventions that can be delivered in routine practice without specialist training are therefore needed. This study evaluates the Happiness Toolkit (HTK), a structured intervention to support early identification and strengthen child self-regulation and family well-being in children screening positive for SEMHD.

The HTK: components and rationale

The HTK comprises six evidence-based techniques delivered by families for 5 min each morning and evening. Each component is described below as it was administered in clinic; component-level administration and uptake are summarised in table 1 .

The neurobiological bases underpinning these components, including dopaminergic, serotonergic and oxytocin pathways, have been described by Sawant et al. 15 Regular practice of the HTK components activates key neurotransmitter pathways, supporting integration between logical and emotional processing and facilitating the shift from reactive to reflective responding. 14 15

Components were selected on the basis of feasibility in routine clinical practice, family acceptability and the availability of supporting evidence. The HTK is adaptable across ages; for younger children, the focus is on guiding parental caregiving. The toolkit is delivered as a single illustrated paper handout (online supplemental file 4) and requires no application, device or specialist equipment. A prior pilot study by Sharif et al (n=20) found that 90% of participants reported symptom improvement at 4–6-week follow-up, with none reporting deterioration, 16 motivating the present larger evaluation.

Study aims

This study aimed to (1) evaluate the feasibility of delivering the HTK in routine paediatric outpatient care; (2) assess parental acceptability and satisfaction and (3) examine preliminary clinical outcomes at 6 weeks and 6 months.

Methods

Study design

Single-site prospective cohort study with prespecified feasibility benchmarks: consent rate 70% or higher, positive screen rate consistent with population estimates and 6-week follow-up response rate 60% or higher.

The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) checklist for cohort studies has been completed and is submitted as online supplemental file 5 .

Setting

Conducted in the general paediatric outpatient department, Midlands Regional Hospital, Mullingar, Ireland, across 16 clinics from 11 June to 24 September 2025.

Participants

All consecutive eligible families attending during the study period were approached prior to consultation in a dedicated clinic room by one or two members of the research team (eligibility for approach: age 0–16 years; parent or guardian present and able to provide written informed consent and access email). Families received plain-language verbal and written information about the HTK and study and time to ask questions (online supplemental file 1). Those who wished to participate provided written informed consent (online supplemental file 2) and, where appropriate, age-appropriate assent.

Following consent, families completed the age-appropriate screening questionnaire. Children screening positive on either instrument, or scoring close to established thresholds at the clinician's discretion, were enrolled and received the HTK. Families screening negative were not enrolled but received the standardised HTK handout as a low-intensity resource.

Screening instruments

Ages and Stages Questionnaires: Social-Emotional, Second Edition

The Ages and Stages Questionnaires: Social-Emotional, Second Edition (ASQ:SE-2) was used for children aged 0–5 years. This parent-completed questionnaire has demonstrated acceptable sensitivity and specificity for social-emotional difficulties in early childhood. 17 Age-specific forms were administered according to the child's age in months, with validated cut-off scores applied.

Screen for Child Anxiety Related Emotional Disorders

The Screen for Child Anxiety Related Emotional Disorders (SCARED) was used for children aged 6–16 years: a 41-item parent-completed and child-completed instrument covering five anxiety domains with validated cut-off scores applied. 18 19 It does not screen comprehensively for all dimensions of SEMHD, particularly conduct difficulties and depressive symptoms; positive screen rates for non-anxiety presentations may therefore be underestimated. The Paediatric Symptom Checklist was not used; this is acknowledged as a limitation. Five children who scored marginally below established screening thresholds were enrolled at the attending clinician's discretion based on clinical presentation.

Intervention delivery

For enrolled families, delivery occurred in two steps. Before consultation, the research team member explained the six HTK components using the standardised handout (online supplemental file 4) and coached families on the 5 min morning and evening practice routine; this took approximately 10–15 min. The treating clinician then reiterated and integrated the HTK within the outpatient consultation, requiring approximately five additional minutes. Total staff contact per enrolled family was approximately 25–30 min.

Follow-up and data collection

Data were collected at baseline in person, 6 weeks and 6 months via secure web-based surveys. Each participating family received a unique pseudonymised study identifier by email; the corresponding linking key, held on paper forms completed at the index clinic visit, was stored in a locked office at Midlands Regional Hospital accessible only to named investigators employed there. No names or directly identifying information were entered into a computer or web-based application at any time point. Non-responders received two email reminders and one telephone follow-up call. Follow-up surveys were completed independently by parents at home without researcher presence, reducing social desirability bias. However, response bias toward families who engaged more positively with the toolkit cannot be excluded. The parent-reported functional outcome survey was developed specifically for this study; no validated parent-report symptom instrument was administered at follow-up.

Child age, sex, neurodevelopmental diagnosis or professional opinion and prior psychology or CAMHS involvement were not collected at enrolment; these variables were included in the 6-week follow-up survey and are therefore available only for respondents at that time point (n=46). Follow-up surveys assessed: adherence (frequency of use, components used); acceptability (ease of implementation, satisfaction, overall experience, willingness to recommend, support for wider implementation); parent-reported changes in child mood, emotion management, emotional expression, sleep, social interaction and screen time on 5-point Likert scales; parental confidence in managing SEMHD; adverse events; and optional free-text qualitative feedback.

Outcome measures

Primary feasibility outcomes were consent rate, positive screen rate and 6-week follow-up response rate; primary acceptability outcome was parental willingness to recommend the HTK. Secondary outcomes were parent-reported functional changes, component uptake, qualitative themes and adverse events.

Safety-netting and clinical governance

Where positive screening was accompanied by clinical features indicating specialist review, the attending paediatrician initiated a psychology and/or CAMHS referral. Referral decisions were based on clinical assessment of symptom severity, functional impairment and existing supports; validated screening scores informed but did not solely determine these decisions. Acute or high-risk presentations prompted same-day senior review and local safeguarding protocols. All families received verbal red-flag safety advice and dedicated written contact details for the paediatric research team.

Statistical and qualitative analysis

Descriptive statistics were used throughout; no formal power calculation was performed. Data are summarised as mean (SD) or median (range) for numerical variables and as frequency (percentage) for categorical variables. Free-text responses were analysed using inductive content analysis; two researchers (KE and SAM) coded independently with discrepancies resolved by consensus.

Analysis followed the qualitative content analysis process described by Elo and Kyngäs, 20 applied post hoc. Responses were exported verbatim and read for familiarisation, then coded openly line by line by both researchers independently; codes were grouped into categories by consensus, with categories derived from the data rather than specified in advance. No qualitative analysis software was used. As the data were written survey responses completed by parents at home, no interviews, recordings or field notes were generated, and neither data saturation nor participant validation of categories was assessed. Reporting follows the Consolidated Criteria for Reporting Qualitative Research (COREQ) 32-item checklist, annotated where individual items do not apply to free-text survey data (online supplemental file 6). 21

Patient and public involvement

Patients and members of the public were not involved in setting the research question, study design, choice of outcome measures, recruitment or dissemination planning for this study. The study was designed by the clinical research team in response to observed unmet need in routine paediatric outpatient practice. This was a pragmatic feasibility study embedded within an existing clinical service; formal patient or public involvement was not undertaken prior to commencement.

Participating families were informed at enrolment that results would be shared in published form. Findings will be disseminated to families and relevant community groups through a lay summary available via the outpatient department at Midlands Regional Hospital Mullingar on publication. Future iterations of the HTK should involve parents and children as partners at the design stage, including those with neurodevelopmental difficulties, who represented a substantial proportion of this cohort.

Results

Participant flow and characteristics

The study was conducted across 16 outpatient clinics from 11 June to 24 September 2025. Of 277 families approached, 240 consented (consent rate 86.6%), meeting the prespecified benchmark of 70% or higher. Following age-appropriate screening, 94 children (39.2% of consented participants) were enrolled and received the HTK. Of these children, 64 screened positive on total SCARED score (parent or child score of 25 or above), 25 on ASQ:SE-2 age-specific cutoffs and 5 were enrolled on clinical grounds while also scoring close to established thresholds. The consent-rate and positive-screen-rate feasibility benchmarks were met; the pre-specified 6-week follow-up response-rate target of 60% was not met (48.9%). A participant flow diagram is presented in figure 1 .

6-week follow-up

Response rate

46 of 94 families (48.9%) completed the 6-week survey, below the prespecified target of 60% despite two email reminders and a telephone reminder call. One duplicate submission was excluded. Among respondents, median child age was 9.5 years (range 4 months to 16); 50.0% were male (n=23) and 50.0% were female (n=23). Of respondents, 45.7% had a prior neurodevelopmental diagnosis or

professional opinion, 34.8% did not and 19.6% were awaiting assessment. 37.0% were linked with formal psychology supports and/or CAMHS or on a waiting list. Characteristics of 6-week follow-up respondents are presented in table 2 .

Among the 21 children with a prior neurodevelopmental diagnosis or professional opinion, autism spectrum disorder was the most commonly reported condition (19 of 21 responses with diagnostic detail), frequently in combination with co-occurring diagnoses including ADHD (n=6), sensory processing disorder (n=5), anxiety disorder (n=3) and dyspraxia (n=3).

Adherence and component utilisation

Frequency of use: once daily (30.4%), twice daily (6.5%), every other day (23.9%), weekly (23.9%) and occasionally (10.9%). Most engaged as a parent-and-child pair; four children used the toolkit independently. Mindfulness and deep breathing (76.1%), visualisation (69.6%) and positive physical contact (65.2%) were most used (table 1 and figure 2).

Acceptability at 6 weeks

95.2% reported a positive or very positive overall experience; 92.9% were very satisfied with the clinical explanation; and 92.9% supported wider implementation in outpatient clinics. Mean ease of implementation was 4.47 on a 5-point Likert scale (SD 0.97). 95.7% reported the HTK helpful and 97.8% would recommend it (table 3).

Parent-reported outcomes at 6 weeks

Parents reported improvements across all functional domains at 6 weeks using 5-point Likert scales; full results are in table 4 and figure 3 . Mood (73.9% substantial improvement) and parental confidence in supporting the child’s emotional needs (80.4%) showed the highest improvement rates. Notably, 43.5% reported reduced screen time and 37.0% reported improved sleep, despite neither being an explicit target of the intervention. These findings likely reflect improved self-regulation rather than adherence to specific behavioural advice, which was not a component of the HTK intervention.

Adverse events at 6 weeks

No adverse events related to the HTK were reported at 6-week follow-up.

Six-month follow-up

The 6-month sample (n=13, 13.8% of enrolled participants) is small. All 6-month findings are exploratory and should not be used to draw generalisable conclusions about sustained efficacy.

Response rate

13 families (13.8% of enrolled participants) completed the 6-month follow-up survey. Enrolled parents received two email reminders and one telephone reminder call. Among respondents, median child age was 12.0 years (range 4–17 years); 69.2% were male (n=9) and 30.8% were female (n=4). 76.9% had a prior neurodevelopmental diagnosis or professional opinion and 23.1% did not; none were awaiting assessment. 53.8% were linked with formal psychology or CAMHS or on a waiting list. One respondent requested exclusion from future data collection and was removed from further follow-up contact.

Compared with 6-week respondents, 6-month completers had a higher proportion with prior neurodevelopmental diagnosis (76.9% vs 45.7%). Non-response may reflect symptomatic improvement or greater clinical complexity limiting engagement; both interpretations are plausible. These comparisons should be interpreted with caution given the small sample.

Adherence and component utilisation at 6 months

Practice frequency declined but persisted in most families. Positive physical contact became the most used component (76.9%), followed by mindfulness (69.2%) and visualisation (46.2%), while the gratitude journal declined markedly (15.4%) (table 1).

Acceptability at 6 months

Mean ease of implementation was 4.31 (SD 0.63); 92.3% rated it easy or very easy; 69.2% reported a positive overall experience; 84.6% supported wider implementation.

Parent-reported outcomes at 6 months

Outcomes at 6 months remained in the improvement range across all domains, with some attenuation compared with 6 weeks (table 4). Mood improvement was reported by 61.5% (mean 3.62, SD 0.51); parental confidence in supporting the child's emotional needs remained high (69.2%; mean 3.92, SD 0.95). 92.3% would recommend the toolkit and 69.2% found it helpful.

Adverse events at 6 months

No adverse events related to the HTK were reported at 6-month follow-up.

Qualitative findings

Content analysis of free-text responses identified three consistent themes

Emotional regulation and expression. Parents consistently described concrete improvements in their child's ability to manage and articulate emotions. Examples included independent use of deep breathing during anxious episodes, reduction in meltdowns and improved ability to verbalise feelings. One parent of an 11-year-old boy with autism, anxiety and developmental language disorder wrote: "He was afraid to go to sleep due to a fear of dying and since introducing the toolkit as part of a bedtime routine he can now fall asleep so much easier." A parent of an 8-year-old girl with autism described replacing self-injurious behaviour with the toolkit's positive touch component: "We have started encouraging her to squeeze one of us in a big bear hug instead of self hitting and it's really made a difference." A parent of a 14-year-old boy described his independent adoption of the toolkit: "Noticed when he was getting angry playing games he will now take a moment and step away and practice some of the steps from HTK. Along with this he is now communicating his emotions as they happen rather than waiting until afterwards and internalising every negative feeling."

Normalisation of positive family practices. Several parents described the HTK as a catalyst for wider family change. One parent wrote that it restored calm and opened communication; another described it as helping her family through their most difficult period, noting her son's dark thoughts about himself and school were no longer present. Multigenerational uptake was noted; one parent reported her daughter's 13-year-old sibling began practising the toolkit independently after observing it being used.

Barriers, facilitators and recommendations. The most commonly reported barrier was establishing initial routine during family stress or where children had significant neurodevelopmental needs. The HTK was designed as a brief paper-based handout with the aim of reducing screen time and moving children towards embodied, relational practices. The standardised handout was central to sustained engagement; multiple parents independently requested laminated versions for durability. One parent of a 15-year-old reflected that ‘maybe try doing it at a younger age’; another adapted the toolkit for her 14-year-old, noting that ‘journaling and conversation works better’ and that older children felt ‘too cool’ for some components. One parent of a child with school avoidance and sertraline-treated anxiety observed that her daughter found the toolkit difficult to use during periods of poor mental health. Families suggested wider access through schools, early intervention services and digital formats.

Discussion

Of 277 families approached, 240 consented (86.6%) and 94 children (39.2%) received the HTK. The consent-rate and positive-screen-rate feasibility benchmarks were met; however, the pre-specified 6-week follow-up response-rate target of 60% was not met (48.9%). At 6-week follow-up (n=46), 95.2% reported a positive experience, 97.8% would recommend and no adverse events were recorded. At 6-month follow-up (n=13, exploratory), 61.5% reported sustained mood improvement and 92.3% would still recommend. The 39.2% positive screening rate in an unselected outpatient population indicates a substantial burden of unmet psychosocial need in routine paediatric care.

Strengths

Strengths include a pragmatic prospective design, validated age-specific screening instruments, an unselected paediatric clinic population and embedded safety netting. Screening was carried out consecutively rather than by referral, follow-up was completed by parents at home without researcher presence, and free-text responses were coded independently by two researchers.

Limitations

The non-randomised single-arm design precludes causal attribution and the 6-week response rate of 48.9% fell below the prespecified target of 60%, introducing potential response bias toward families who engaged more positively with the toolkit. Demographic and neurodevelopmental data were not collected at enrolment and are available only for 6-week follow-up respondents (n=46), limiting characterisation of the full cohort. The 6-month sample is small and all 6-month findings are exploratory. The single-site design limits generalisability and data collection was carried out by voluntary clinical staff and medical students alongside routine duties; future evaluations should incorporate dedicated research staffing.

The screening strategy was also necessarily pragmatic. The SCARED is a validated and widely used tool and anxiety is the most common mental health condition in children, but it does not comprehensively identify other conditions such as ADHD, autism spectrum disorder or broader behavioural and depressive difficulties. Instruments such as the Paediatric Symptom Checklist and the SNAP-IV for ADHD 22 23 would have captured a wider range of presentations, but administering multiple tools was not feasible within the time available to volunteer researchers in a running outpatient clinic. The prevalence of difficulty reported here may therefore be an underestimate, and future studies with dedicated research support should incorporate a broader screening panel.

Comparison with previous work

These findings replicate the prior conference abstract by Sharif et al, 16 in which 90% of participants reported symptom improvement at 4–6 weeks. 16 The neurobiological basis for the toolkit components has been described by Sawant et al , who identified dopaminergic, serotonergic and oxytocin-mediated mechanisms, and vagal activation via breathing, as key mediators. 15 By incorporating these components into daily practice, neuronal pathways are activated supporting integration between logical and emotional processing and facilitating the shift from reactive to reflective responding, 14 potentially building resilience over time. The rise in positive physical contact from 6 weeks to 6 months (65.2%–76.9%) corresponds with medium-sized effects on anxiety, depression and stress in the touch intervention literature. 13 Notably, 37.0% of parents reported improved sleep and 43.5% noted reduced screen time, neither an explicit target of the intervention. The sleep finding may reflect downstream serotonergic effects of the mindfulness, breathing and positive touch components; sleep disturbance is common in children with SEMHD and even modest improvements have significant clinical and family consequences.

Clinical and policy relevance

The clinical relevance of this study lies in demonstrating that a low-cost, family-delivered intervention can be integrated into existing paediatric outpatient practice without additional specialist staffing or equipment. In a health system where 4462 children await CAMHS and over 28 000 await primary care psychology, the HTK provides a practical means of support during what is currently an unstructured waiting period. One parent described it directly: “She is using the toolkit awaiting CAMHS.” This positions the HTK within a stepped-care model, complementing rather than replacing specialist services when subsequently accessed. The qualitative data demonstrate use across a wide clinical spectrum, from a non-verbal child with autism to a child with sertraline-treated school avoidance. Autism spectrum disorder was the most frequently reported condition in this cohort, often with co-occurring ADHD, sensory processing difficulties and anxiety, suggesting the toolkit is adaptable across neurodevelopmental complexity. A brief, clinician-directed, family-delivered approach achieving high consent rates and strong acceptability in an unselected population has clear implications for early intervention policy in paediatric mental health.

Future research

A cluster randomised controlled trial with validated symptom outcomes and economic modelling is warranted. Future iterations should incorporate booster sessions, formal patient and public involvement at the design stage and age-appropriate or digital adaptations for older adolescents. The screen time reduction and sleep improvement findings are exploratory but warrant prospective investigation given their potential to reflect broader self-regulation benefit. The neurobiological mechanisms described by Sawant et al 15 suggest that serotonergic and dopaminergic pathways may mediate these effects and should be examined in future work.

Conclusions

The HTK can be feasibly implemented in routine paediatric outpatient care with high acceptability and no adverse events at two time points. 86.6% of families consented, 39.2% screened positive for SEMHD and 97.8% of 6-week respondents would recommend the toolkit. Findings must be interpreted in the context of an uncontrolled design, a response rate below target, a small exploratory 6-month sample and the absence of a dedicated research team; causal attribution is not possible. The high consent rate, strong acceptability and absence of harm support progression to a controlled evaluation.

In Ireland, 4462 children await CAMHS assessment and over 28 000 await primary care psychology services, 6 7 with waits of up to 10 years for primary care psychology in some areas. The HTK can be delivered without specialist input during the period families await formal assessment, providing structured early support at a time when none currently exists.

Supplementary material